(866) 472-6723 Email

MAHARISHI SCHOOL

STEM Strength

INNOVATE AND LEAD

Building the next generation of innovators

Our STEM Strength comes through our unparalleled opportunities for students to push beyond the traditional classroom and immerse themselves in cutting-edge fields. Through dynamic, hands-on experiences in **Robotics**, **Aviation**, and the prestigious **NASA Tech Rise Student Challenge**, students can explore their passions and build a powerful foundation for the future.

These programs are designed to challenge participants to apply complex concepts to real-world problems, whether they're engineering a competitive robot, mastering the principles of flight, or designing an experiment destined for suborbital space research. By tackling these ambitious projects, students not only sharpen their **tech**

and engineering skills but also develop crucial life skills in **leadership**, **collaboration**, **and critical thinking**, preparing them to become the next generation of innovators.

ENGINEERING | PLANNING | LEADERSHIP

Robotics

Our Upper School Robotics program is an immersive, hands-on experience where students do more than just build a robot—they will run a complete team. This multifaceted program blends engineering, programming, business, and community engagement to provide a comprehensive foundation for a future in STEM. They will gain real-world business acumen by managing a budget, fundraising through professional pitches to local sponsors, and acting as an ambassador for STEM in the community. These efforts in outreach and resource development are crucial, as they form the bedrock for a successful team and are essential for earning prestigious honors like the Connect Award at competitions.

The technical heart of the program lies in bringing a competitive robot to life from the ground up. They will collaborate to design and construct the robot’s chassis and drivetrain, then engineer complex sub-systems like arms, intakes, and launchers to perform specific game tasks, all while ensuring reliability under pressure. Before any physical building begins, they will master industry-standard tools like Fusion 360 to create detailed 3D models and run simulations in a virtual environment, a process key to winning the Design Award.

The robot’s intelligence comes from its code, as it masters Java to program both autonomous and driver-controlled modes, striving for the coding excellence required to compete for the Control Award.

Documenting this entire journey is as critical as the build itself. The team will meticulously maintain an Engineering Notebook, an official record that chronicles the design process with sketches, calculations, and team decisions. This notebook also requires you to articulate the real-world applications of the science and engineering principles used in your robot’s design. A detailed and professional portfolio of their work is not only a key reason teams advance in competition but is often a mandatory component for the highest awards, solidifying the program’s ethos that success is built on a fusion of brilliant ideas, skilled execution, and thorough documentation.

PILOT | AIRSPACE | FLIGHT TRAINING

Aviation

Students gain a strong foundation in the fundamentals of flight, aircraft systems, navigation, weather, and aviation regulations—essential knowledge for anyone considering student pilot training or a future career in aviation.

The course includes classroom instruction, hands-on activities, and an unforgettable **Discovery Flight** with a licensed, experienced flight instructor. During their introductory flight, each student will have the opportunity to take off, fly, and land an aircraft under professional supervision—an inspiring first step toward earning a pilot's license.

Students will be introduced to the topics covered on the FAA's Private Pilot Ground School exam, including:

- Principles of flight and aerodynamics
  Aircraft instruments and systems
- Airspace and flight planning
- Meteorology and weather interpretation
- FAA regulations and air traffic control
  Aviation decision-making and safety

By the end of the course, students will:

- Understand the basic concepts of manned and unmanned flight
- Experience hands-on flying with both airplanes and drones
- Be prepared to begin formal Student Pilot ground school training
- Hold an FAA Recreational Drone License

This project is an Introduction to Aviation fundamentals for high school students to explore the exciting, growing field of aviation!

NASA | ENGINEERING | COMPETITION

Tech Rise

This is a unique opportunity to design, build, and launch a real science experiment on a high-altitude balloon that will travel to the stratosphere.

The student team's mission is to investigate one of the most significant challenges in space exploration: Radiation.

This project is directly inspired by the need to protect humans from the harmful effects of radiation, which is abundant in space and a major obstacle for long-duration spaceflight.

What sets this experience apart is the direct connection to the space industry. Our team will have **weekly meetings with NASA Outreach Engineers**, receiving expert guidance and mentorship throughout the entire project lifecycle. You will learn invaluable engineering, project management, and collaboration skills while contributing to real-world scientific research. Join us to turn a hypothesis into a high-flying experiment!

INQUIRE TODAY!

Igniting the Next Generation of Innovators

At the heart of our educational philosophy lies a commitment to transforming students from passive learners into active creators and problem-solvers. Our dynamic suite of STEM programs is designed to do just that, providing hands-on, minds-on experiences that extend far beyond the textbook. In recognition of this excellence, we have been ranked **#1 in STEM by Niche.com**—a testament to our unparalleled commitment to preparing students for the future.

We move beyond theory to immerse students in the real-world challenges of Rocketry, Robotics, NASA TechRise, Aviation, and competitive Science Fairs. Each project is a launchpad for developing the essential skills that define tomorrow's leaders.

Our Pillars of Excellence

Every project is built upon a foundation of core goals that prepare students for success in any field:

Creativity & Critical Thinking: We challenge students to design novel solutions, whether it's programming a robot to navigate a complex obstacle or designing a rocket's payload for a unique mission. There is no single "right answer," only the best solution they can engineer.

Collaboration & Communication: Our students learn to work as a cohesive team, mirroring professional engineering and research environments. They articulate their ideas, delegate tasks, and present their findings with clarity and confidence to peers and judges.

A Growth Mindset & Interdisciplinary Integration: We celebrate "failure" as a vital step in the learning process. A rocket that doesn't launch perfectly or a code that fails is an opportunity to learn, iterate, and improve. Our programs naturally blend physics, mathematics, computer science, and design, showing students how knowledge connects in the real world.

Societal Impact & Service: We empower students to see themselves as agents of change. From developing robotics solutions for community challenges to designing aviation tech for environmental monitoring or conducting research with real-world implications for Science Fairs, our students learn that STEM is a powerful tool for service.

Our Mission

To create an innovative Consciousness-Based educational environment where students think deeply and become creative, compassionate, contributing citizens of the world.

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